Exp.no:8 NMAP to Discover Live Host using Nmap Scans

Aim:

To discover live hosts on a network using NMAP host discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of TCP/IP Model and when each one applies.

1. ARP Host Discovery

The ARP scan successfully identified the active hosts in the local subnet. ARP request were broadcast to the network and the live hosts responded with their MAC address.

This confirms that ARP based discovery works at the Link layer and is mainly used for discovering hosts on the same local network.

2. ICMP Echo scan (-PE)

The ICMP Echo scan identified hosts that responded to ICMP Echo Requests. The hosts that replied were considered Active.

3. ICMP Timestamp Scan (-PP)

The ICMP Timestamp scan identified some live hosts that responded to timestamp requests some hosts didn't responded because ICMP timestamp packets may be filter by firewalls.

4. ICMP Address Mask scan (-PM)

The ICMP Address Mark Scan received few or no responses. This is because address mask requests are commonly blocked or unsupported by modern systems and firewalls.

5. TCP SYN Ping (-PS)

The TCP SYN ping identified active hosts by sending TCP SYN packets to the specified port.

An SYN/ACK response from an open port or an RST response from a closed port or an RST Response.

6. TCP ACK Ping (-PA)

The TCP ACK Ping identified active hosts by sending TCP packets with the ACK flag set. send the packets ~~with the~~ does not belong to an existing connection, a live host generally responds with an RST packet.

7. UDP Ping:

The UDP ping identified active hosts by sending UDP packets to the specified port.

When the target port is closed, the host normally responds when an ICMP port unreachable message confirming that hosts is active.

Result:

The NMAP host discovery experiments were successfully performed using ARP, ICMP, TCP & UDP scanning techniques.